

PEM-CAN: Parameter-Efficient Multimodal Fusion for Diabetic Neuropathy Screening

Seyed Mahmoud Sajjadi Mohammadabadi^{a,*}

^aDepartment of Computer Science and Engineering, University of Nevada, Reno, Reno, NV, 89557, USA

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ABSTRACT

Multimodal foundation models for ambulatory screening face three core engineering bottlenecks: quadratic attention memory expansion $\mathcal{O}((M+L)^2)$ over multi-day biosignal streams, visual modality dominance over low-amplitude autonomic signals, and severe telemetry packet loss. We formulate the Parameter-Efficient Multimodal Cross-Attention Network (PEM-CAN), an architecture that freezes pre-trained visual foundation encoders (ViT-B/16, 86.6 M parameters) and confines cross-modal learning to low-rank adapter projections ($r = 8$) regularized by a Stiefel-manifold orthogonality penalty ($\mathcal{L}_{\text{orth}} = \|AA^T - I_r\|_F^2$). Evaluated on a metabolic-autonomic cohort statistically mapped to the NIH Bridge2AI AI-READI Type 2 Diabetes protocol schema ($N = 2,840$ patient profiles; fundus photography, Dexcom G6 continuous glucose monitoring with $T_g = 2,016$ steps, and Garmin Vivosmart 5 wearable actigraphy across 5 channels), PEM-CAN achieves 0.934 AUROC [95% CI: 0.904, 0.960], an AUPRC of 0.795 ± 0.011 , an F1-score of 0.803 ± 0.012 , and an Expected Calibration Error of 0.038 at 90.5% accuracy on the held-out test split ($N = 568$, 22.5% prevalence), bounded below the theoretical Bayes optimal oracle ceiling of 0.957 AUROC. Under 50% contiguous telemetry dropout, the model retains 0.861 AUROC (92.2% baseline retention). On a synthetic cognitive-autonomic cohort fusing heart rate variability, electroencephalography band powers, actigraphy, and speech features, PEM-CAN recovers non-linear interaction terms at 0.942 ± 0.007 AUROC under adequate sample size ($N = 1,440$), while sample starvation ($N = 360$) degrades discrimination to 0.854 ± 0.015 due to covariance estimation variance. The architecture updates 2.60 M parameters (1.75% of the 148.6 M model), reduces peak training VRAM from 32.4 GB to 8.4 GB, executes at 20.2 ms latency on NVIDIA Jetson Orin Nano (1.18 GB FP16), and slashes federated client bandwidth by 98.25% (10.4 MB/round) under FFA-LoRA.

1. Introduction

Diabetic autonomic neuropathy (DAN) is among the most clinically devastating yet frequently overlooked microvascular complications of diabetes mellitus, doubling the five-year mortality rate of affected patients through silent myocardial infarction, malignant cardiac arrhythmias, and sudden metabolic decompensation [1, 2, 3]. Despite its lethal trajectory, clinical diagnosis remains largely reactive [4]. Standard staging depends on specialized cardiovascular autonomic reflex tests (CARTs) or sudomotor axon reflex evaluations, which require dedicated autonomic laboratory infrastructure and are rarely performed during routine outpatient visits [5]. Furthermore, conventional glycemic monitoring via glycated hemoglobin (HbA1c) represents a blunt retrospective instrument; by condensing three months of complex metabolic fluctuations into a single blurred mean, HbA1c remains blind to rapid postprandial glucose surges, nocturnal hypoglycemic crashes, and circadian autonomic variability that directly accelerate microvascular and neurovascular damage [12, 13].

To transcend these diagnostic blind spots, biomedical informatics must unify two distinct observational windows into human physiology [16, 14]:

1. **The Ocular Window (Cumulative Structural Damage):** The retina is embryologically and anatomically

an exposed extension of the central nervous system [9]. High-resolution retinal fundus photography non-invasively captures the microvascular architecture of the human body, revealing capillary rarefaction, arteriolar narrowing, and microaneurysms that correlate directly with systemic autonomic and peripheral nerve fiber attrition [6, 7, 8, 10, 11].

2. **The Wearable Window (Dynamic Physiological Provocation):** Continuous glucose monitoring (CGM) and commercial wearable actigraphy capture the dynamic metabolic and autonomic excursions that drive microvascular damage [12, 14]. Wearable sensor telemetry provides longitudinal recordings of interstitial glucose fluctuations, resting heart rate volatility, respiratory patterns, sleep architecture, and autonomic stress scores over days and weeks.

The NIH Bridge2AI **AI-READI** initiative establishes an equitable multi-organ data standard coupling macula- and disc-centered fundus photographs with continuous Dexcom G6 glucose telemetry and Garmin Vivosmart 5 wristband actigraphy across diverse Type 2 Diabetes cohorts [15]. While access to raw clinical repositories is governed by institutional review boards and data use agreements, open and statistically calibrated simulation benchmarks provide a mathematically rigorous testbed to evaluate multimodal fusion architectures under controlled physiological signal-to-noise ratios.

*Corresponding author.

Email address: mahmoud.sajjadi@unr.edu (S.M.S. Mohammadabadi)
ORCID(s): 0009-0001-9629-9734 (S.M.S. Mohammadabadi)

Translating coupled ocular-wearable telemetry into deployable clinical artificial intelligence exposes three critical contextual deployment hurdles [16, 18]:

- **On-Device Memory Budgets and Attention Complexity:** Point-of-care clinical workstations and mobile screening carts operate under constrained GPU memory budgets. Concatenating temporal biosignal tokens ($L > 2,000$) with visual patch tokens ($M = 576$) inside standard self-attention architectures produces quadratic memory expansion $\mathcal{O}((M+L)^2)$ [19]. Full fine-tuning of massive visual transformers during multi-day surveillance training routinely triggers out-of-memory faults.
- **Pervasive Ambulatory Channel Dropout:** In real-world outpatient monitoring, wearable sensors frequently suffer 10% to 50% contiguous packet loss resulting from battery depletion, Bluetooth transmission dropouts, sleep sensor compression, or device non-compliance [18]. Conventional multimodal networks undergo catastrophic performance degradation when time-series channels drop out.
- **Institutional Data Governance and Bandwidth Boundaries:** Centralizing raw retinal photography and wearable telemetry across independent hospital networks violates patient privacy statutes (HIPAA and GDPR). While federated learning enables decentralized training without pooling data [30, 31, 32], transmitting entire foundation model weight updates (> 500 MB/round) exceeds institutional bandwidth allocations.

Existing multimodal fusion paradigms fail to resolve these deployment constraints due to two structural engineering gaps:

- **Early Feature Concatenation Gaps:** Projecting concatenated raw inputs into a shared multilayer perceptron exposes optimization to modality dominance [17, 18]. High-entropy spatial visual gradients overpower low-amplitude physiological temporal signals, causing the network to ignore subtle autonomic volatility. Furthermore, early fusion architectures lack mechanisms to handle missing channels, collapsing when wearable packets drop.
- **Late Decision Fusion Gaps:** Training separate unimodal classifiers and combining their terminal decision probabilities discards fine-grained cross-modal correlations entirely. Late fusion assumes conditional independence between ocular lesions and glycemic volatility, failing to capture complementary mutual information $I(Z_s; Y \mid Z_v)$ that defines subclinical autonomic neuropathy.

To resolve these challenges, we present the **Parameter-Efficient Multimodal Cross-Attention Network (PEM-CAN)**. PEM-CAN freezes pre-trained visual foundation backbones, injecting lightweight low-rank adapters ($r = 8$)

directly into bidirectional cross-attention projections under a Stiefel-manifold orthogonal regularizer [24, 29]. By constraining cross-modal interaction to an orthogonal low-rank subspace, PEM-CAN mitigates modality collapse, scales with linear computational complexity $\mathcal{O}(M \cdot L)$, and maintains stable inference under severe sensor missingness.

The core technical contributions of this study are four-fold:

1. **Foundation Ocular-Wearable Fusion Architecture (Sections 3 and 3.2):** We formulate an architectural framework that unifies high-resolution 2D retinal fundus photography with multi-day 1D continuous glucose monitoring and 5-channel wearable actigraphy, handling cross-modal dimensional and sampling asymmetries while keeping foundation visual backbones frozen.
2. **Stiefel-Manifold Orthogonal Low-Rank Adaptation (Sections 3.3 and 3.4):** We introduce an orthogonal low-rank cross-attention mechanism constrained by a Stiefel-manifold penalty ($\mathcal{L}_{\text{orth}}$), preventing visual gradient dominance, maintaining projection rank, and ensuring strictly linear memory scaling $\mathcal{O}(M \cdot L)$ over multi-week surveillance horizons.
3. **Multi-Cohort Empirical Benchmark Validation (Sections 4.1, 4.3, and 4.7):** We establish comprehensive evaluations across two distinct cohorts: (i) a primary metabolic-autonomic cohort mapped to the NIH Bridge2AI AI-READI Type 2 Diabetes protocol schema ($N = 2,840$), demonstrating superior diagnostic discrimination (0.934 AUROC) strictly bounded below the Bayes optimal oracle ceiling (0.957 AUROC); and (ii) a synthetic cognitive-autonomic interaction cohort (HRV, EEG, actigraphy, speech), quantifying exact sample-size bounds ($N = 1,440$ vs. $N = 360$) and sensor noise ceilings.
4. **Embedded Edge Profiling and Federated Scaling (Sections 4.2 and 4.13):** We benchmark execution on physical hardware (20.2 ms latency, 1.18 GB FP16 footprint on NVIDIA Jetson Orin Nano) and demonstrate up to 99.56% reduction in client transmission bandwidth across 50 decentralized hospital nodes using federated low-rank aggregation.

2. Related Work

2.1. Multimodal Deep Learning in Diabetes and Oculomics

Deep learning has achieved notable success in ophthalmic disease screening, from diabetic retinopathy classification [8, 7] to systemic biomarker prediction from retinal photography [6, 9]. The release of retinal foundation models such as RETFound [10] demonstrated that self-supervised pre-training on large fundus repositories yields transferable representations. Concurrently, continuous glucose monitoring and wearable biosensors have been used to quantify glycemic variability (Time in Range, Mean Amplitude of Glycemic Excursions) and predict metabolic parameters [12,

[13, 14]. However, prior work largely analyzes retinal imaging and physiological biosignals in isolation or through late decision fusion. Joint end-to-end architectures that explicitly correlate spatial microvascular lesions with continuous temporal metabolic stress remain largely unexplored.

2.2. Parameter-Efficient Fine-Tuning (PEFT)

To adapt massive foundation models without full re-training, parameter-efficient fine-tuning (PEFT) methods update small auxiliary modules while freezing backbone weights. Prominent approaches include Low-Rank Adaptation (LoRA) [24], which decomposes weight updates into low-rank factor matrices $\Delta W = BA$, Weight-Decomposed LoRA (DoRA), Bottleneck Adapters [25], BitFit (bias-only tuning) [26], and dynamic-rank allocation (AdaLoRA) [27]. Furthermore, subspace-oriented reparameterization frameworks such as SOLAR [38] demonstrate that exploiting foundation singular subspaces can decouple adapter representations to dramatically reduce communication and storage overhead in distributed systems. In multimodal applications, adapters have primarily been explored for vision-language tasks. PEM-CAN extends low-rank adaptation to bidirectional cross-modal attention between spatial 2D images and 1D longitudinal biosignals, introducing Stiefel-manifold orthogonal constraints [29] to prevent visual modality dominance.

2.3. Missing-Modality Learning and Gated Fusion

Ambulatory wearable biosensors regularly suffer from sensor detachment, motion artifacts, and transmission dropouts [18]. Architectural strategies for handling missing modalities include Gated Multimodal Units (GMU) [22], Multimodal Transfer Modules (MMTM) [23], and cross-modal attention transformers such as MulT [20] and Perceiver IO [21]. However, full fine-tuning of these models incurs substantial memory overhead during multi-day surveillance training. PEM-CAN provides an alternative by pairing frozen unimodal representations with low-rank cross-attention adapters that maintain graceful performance degradation under up to 50% sensor packet loss.

3. System Architecture and Methodological Formulation

3.1. Problem Formulation and Clinical Objective

Let patient screening observations be represented as an asymmetric multimodal tuple $\mathcal{X} = \{x_v, x_g, x_a\}$, where $x_v \in \mathbb{R}^{H_v \times W_v \times C_v}$ denotes a high-resolution 2D retinal fundus photograph ($384 \times 384 \times 3$), $x_g \in \mathbb{R}^{T_g \times 1}$ denotes a 1D continuous glucose monitoring telemetry stream ($T_g = 2,016$ steps spanning 7 days at 5-minute sampling), and $x_a \in \mathbb{R}^{T_a \times C_a}$ represents multi-channel wearable actigraphy ($T_a = 2,016$, $C_a = 5$ channels: resting heart rate, respiratory rate, device-derived stress score, cumulative steps, and sleep stage classification). The diagnostic objective is to estimate the posterior probability distribution $P(\mathcal{Y} \mid \mathcal{X})$ over four

clinically adjudicated diabetic microvascular and neurovascular complications $\mathcal{Y} = [y_{\text{DAN}}, y_{\text{DR}}, y_{\text{CKD}}, y_{\text{DPN}}]^T \in \{0, 1\}^4$.

3.2. Modality-Specific Feature Embedding

To accommodate the dimensional and sampling disparities between spatial photographs and temporal time-series, PEM-CAN projects each input into a unified embedding dimension $d_m = 768$:

1. Visual Tokenization: Retinal fundus images x_v are processed using a pre-trained Vision Transformer (ViT-B/16) [10] parameterized by weights Θ_v . The image is partitioned into non-overlapping patches of resolution $P \times P = 16 \times 16$, yielding $M = (H_v \cdot W_v)/P^2 = 576$ visual patch tokens:

$$Z_v^{(0)} = [x_v^1 E_v; x_v^2 E_v; \dots; x_v^M E_v] + E_{pos}^{(v)} \in \mathbb{R}^{M \times d_m}, \quad (1)$$

where $E_v \in \mathbb{R}^{(P^2 C_v) \times d_m}$ is the patch projection matrix and $E_{pos}^{(v)} \in \mathbb{R}^{M \times d_m}$ represents 1D learnable position embeddings. The foundation visual encoder weights Θ_v remain strictly frozen during all training phases ($\nabla_{\Theta_v} \mathcal{L} = 0$).

2. Continuous Glucose Embedding: Glucose streams x_g undergo multi-scale 1D temporal convolution with kernel sizes $k \in \{3, 7, 15\}$ to capture both rapid postprandial glycemic excursions and circadian baselines. Convolved features are projected to dimension d_m :

$$Z_g = \text{LayerNorm}(\text{Conv1D}_{\text{multi}}(x_g)W_g + E_{pos}^{(g)}) \in \mathbb{R}^{L_g \times d_m}, \quad (2)$$

where $L_g = 252$ represents temporal tokens subsampled via stride-8 pooling.

3. Wearable Actigraphy Embedding: The 5-channel wearable matrix x_a is embedded using a 3-layer dilated Temporal Convolutional Network (TCN) with residual connections:

$$Z_a = \text{TCN}(x_a; \Theta_a) \in \mathbb{R}^{L_a \times d_m}, \quad L_a = 252. \quad (3)$$

The temporal physiological tokens are concatenated into a joint sensor sequence:

$$Z_s = [Z_g \parallel Z_a] \in \mathbb{R}^{L \times d_m}, \quad L = L_g + L_a = 504. \quad (4)$$

3.3. Parameter-Efficient Subspace Adaptation

Rather than updating the full weight matrices $W_0 \in \mathbb{R}^{d_m \times d_m}$ across cross-attention projections, PEM-CAN injects low-rank factor adapters [24]:

$$W = W_0 + \Delta W = W_0 + \frac{\alpha}{r} BA, \quad (5)$$

where $B \in \mathbb{R}^{d_m \times r}$ and $A \in \mathbb{R}^{r \times d_m}$ with adapter rank $r \ll d_m$ ($r = 8$), and α is a scaling hyperparameter set to $\alpha = 16$. Matrix A is initialized via random Gaussian initialization $\mathcal{N}(0, \sigma^2)$ and subsequently orthogonalized via QR decomposition; matrix B is initialized to zero, ensuring $\Delta W = 0$ at the onset of training.

To prevent cross-modal representation collapse and ensure that visual gradients do not overwhelm low-amplitude physiological representations, we enforce an explicit Stiefel-manifold orthogonality penalty on the factor projections:

$$\mathcal{L}_{\text{orth}} = \|AA^\top - I_r\|_F^2 = \sum_{i=1}^r \sum_{j=1}^r (\langle a_i, a_j \rangle - \delta_{ij})^2, \quad (6)$$

where $\|\cdot\|_F$ denotes the Frobenius norm, $a_i \in \mathbb{R}^{d_m}$ is the i -th row of A , and I_r is the $r \times r$ identity matrix. Constraining $A \in \text{St}(r, d_m)$ ensures that adapter basis vectors span mutually orthogonal directions, preserving maximum projection entropy.

3.4. Bidirectional Cross-Modal Attention and Optimization

Directional interactions between spatial retinal tokens Z_v and physiological sensor tokens Z_s are formalized through bidirectional cross-attention:

$$Q_v = Z_v W_Q^{(v)}, \quad K_s = Z_s W_K^{(s)}, \quad V_s = Z_s W_V^{(s)}, \quad (7)$$

$$S(Z_v, Z_s) = \text{Softmax} \left(\frac{Q_v K_s^\top}{\sqrt{d_k}} \right) V_s, \quad (8)$$

where $H = 8$ attention heads are used with per-head dimension $d_k = d_m/H = 96$. Crucially, cross-attention complexity is $\mathcal{O}(M \cdot L)$, which scales strictly linearly with sensor token length L for a fixed image resolution ($M = 576$), avoiding the quadratic $\mathcal{O}((M + L)^2)$ blowup of joint self-attention.

The reciprocal sensor-to-visual representation $S(Z_s, Z_v)$ is computed symmetrically. Sequence outputs are pooled and concatenated:

$$\bar{z} = [\text{AvgPool}(S(Z_v, Z_s)) \parallel \text{AvgPool}(S(Z_s, Z_v))] \in \mathbb{R}^{2d_m}. \quad (9)$$

Global mean pooled modality representations are defined as $\bar{Z}_v = \frac{1}{M} \sum_{i=1}^M Z_{v,i}$ and $\bar{Z}_s = \frac{1}{L} \sum_{j=1}^L Z_{s,j}$. The network is optimized end-to-end minimizing a composite loss function:

$$\mathcal{L}_{\text{total}} = \sum_{k=1}^4 \mathcal{L}_{\text{focal}}(y_k, \hat{y}_k) + \lambda_1 \mathcal{L}_{\text{orth}} + \lambda_2 \mathcal{L}_{\text{align}}, \quad (10)$$

where $\mathcal{L}_{\text{align}} = 1 - \frac{\bar{Z}_v \cdot \bar{Z}_s}{\|\bar{Z}_v\| \|\bar{Z}_s\|}$ penalizes angular divergence between global modality pooled representations, with $\lambda_1 = 10^{-3}$ and $\lambda_2 = 10^{-2}$.

4. Experimental Design and Results

4.1. Calibrated Simulation Benchmark and Generative Model

To evaluate multimodal fusion under controlled physiological distributions without violating institutional patient

data governance, we formulate two statistically calibrated simulation cohorts: a primary **Metabolic-Autonomic Cohort** mapped directly to the public NIH Bridge2AI AI-READI schema [15], and a secondary **Synthetic Cognitive-Autonomic Cohort** designed to stress-test non-linear cross-modal interaction recovery under varying sample densities and noise regimes.

4.1.1. Primary Metabolic-Autonomic Cohort (AI-READI Schema, $N = 2,840$)

All simulated profiles in this cohort represent diagnosed Type 2 Diabetes patients across varying severity strata:

1. **Latent Physiological Disease Profile:** Each subject i receives a 4-dimensional latent profile $\mathbf{z}_i = [z_{v,i}, z_{g,i}, z_{a,i}, z_{c,i}]^\top \sim \mathcal{N}(\mathbf{0}, \Sigma_{\text{latent}})$, parameterizing retinal microvascular damage (z_v), glycemic volatility (z_g), autonomic neuropathy (z_a), and circadian disruption (z_c):
2. **Demographics and Clinical Covariates:** Covariate marginals match AI-READI population proportions: Non-Hispanic White (60%), Black / African American (25%), and Hispanic / Latino (15%). Age is generated as $52 + 11\epsilon_{\text{age}} + 2z_{v,i}$, diabetes duration as $4 + 3z_{g,i} + \text{Exp}(1/4)$, HbA1c as $6.8 + 0.9z_{g,i} + \mathcal{N}(0, 1.2^2)$, BMI as $28 + 1.2z_{g,i} + \mathcal{N}(0, 4.5^2)$, and systolic blood pressure as $125 + 5z_{v,i} + \mathcal{N}(0, 14^2)$.
3. **Retinal Fundus Embeddings and Pigmentation Stress Test:** Spatial token arrays $Z_v \in \mathbb{R}^{M \times d_m}$ ($M = 576, d_m = 768$) model patch representations extracted by frozen ViT-B/16 from 384×384 photographs. Optical reflectance contrast is attenuated by 10% in Black and Hispanic profiles to simulate choroidal melanin absorption, normalized by adaptive histogram equalization (CLAHE).
4. **Continuous Glucose Telemetry:** 7-day CGM series $x_g \in \mathbb{R}^{2016 \times 1}$ (Dexcom G6 5-minute sampling) are generated with baseline glucose $105 + 35z_{g,i}$ mg/dL, circadian rhythm $12 \sin(\omega t)(1 + 0.3z_{c,i})$, and stochastic volatility $\sigma_i = 14 + 18z_{g,i}$ mg/dL, standardized as $(x_g - 120)/40$.
5. **Wearable Actigraphy Schema:** Telemetry $x_a \in \mathbb{R}^{2016 \times 5}$ follows the Garmin Vivosmart 5 schema (5 channels: resting heart rate, respiration rate, device stress score, step count, and sleep duration), scaled inversely with autonomic impairment z_a .
6. **Clinical Labels and Prevalences:** Multi-task binary outcomes are assigned via calibrated logistic links:

$$\ell_{\text{DAN}}(\mathbf{z}) = -3.65 + 2.18z_a + 1.42z_g + 1.65z_v + 0.98z_c, \quad (12)$$

$$\ell_{\text{DR}}(\mathbf{z}) = -3.65 + 2.65z_v + 1.25z_g + 0.55z_a, \quad (13)$$

$$\ell_{\text{CKD}}(\mathbf{z}) = -4.18 + 1.65z_v + 1.85z_g + 1.15z_a, \quad (14)$$

$$\ell_{\text{DPN}}(\mathbf{z}) = -2.52 + 2.25z_a + 1.45z_g + 0.85z_v. \quad (15)$$

Observed cohort prevalences match target epidemiological benchmarks: DAN (22.5%), DR (18.2%), CKD (14.1%), and DPN (26.8%). The theoretical **Oracle Bayes Optimal Ceiling** computed directly from ground-truth latent state $\mathbf{z}$ achieves an AUROC of 0.957 [95% CI: 0.935, 0.975] and AUPRC of 0.887 for DAN.

4.1.2. Synthetic Cognitive-Autonomic Cohort

($N = 1,440$ vs. $N = 360$)

To quantify the statistical capacity of orthogonal cross-attention to isolate non-linear cross-modal dependencies independently of retinal imaging, we formulate a synthetic cognitive-autonomic interaction suite:

- **Sensor Modalities:** Continuous telemetry streams encapsulate 4 neuro-autonomic channels: (i) Heart Rate Variability (HRV time-frequency features: LF/HF ratio, RMSSD, SDNN), (ii) Electroencephalography band powers (EEG frontal theta 4 – 8 Hz, alpha 8 – 12 Hz, beta 13 – 30 Hz), (iii) Wrist actigraphy acceleration (triaxial vector magnitude), and (iv) Vocal acoustic features (pitch jitter, shimmer, formant dispersion).
- **Non-Linear Cross-Modal Interaction:** A binary autonomic-cognitive fatigue label is generated via non-linear interaction terms $\text{logit } P(y = 1) = \beta_0 + \sum_j \beta_j x_j + \sum_{j < k} \gamma_{jk}(x_j \cdot x_k) + \epsilon_{\text{noise}}$, where $\epsilon_{\text{noise}} \sim \mathcal{N}(0, \sigma^2)$ with $\sigma \in [0.1, 0.5]$.
- **Sample-Size Strata:** Models are trained under both an adequately powered cohort ($N = 1,440$ instances) and a sample-starved clinical regime ($N = 360$ instances) to evaluate covariance estimation stability.

Clinical Endpoint Definitions:

- **DAN (22.5% prevalence):** Diabetic Autonomic Neuropathy, screened via validated composite autonomic criteria (abnormal heart rate response to deep breathing or COMPASS-31 score ≥ 16 [35]).
- **DR Grade ≥ 2 (18.2% prevalence):** Moderate non-proliferative diabetic retinopathy or worse based on ICDR criteria [8].
- **CKD (14.1% prevalence):** Chronic Kidney Disease defined by KDIGO criteria (eGFR < 60 mL/min/1.73 m² or persistent albuminuria UACR ≥ 30 mg/g) [36].
- **DPN (26.8% prevalence):** Diabetic Peripheral Neuropathy defined by confirmed loss of protective sensation via 10-g monofilament examination or Toronto Clinical Neuropathy Score (TCNS ≥ 6) [37].

Evaluations use 5-fold cross-validation (70% train, 10% validation, 20% held-out test split, $N = 568$ for the metabolic cohort). Models are optimized using AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay 10^{-4}), initial learning rate 10^{-3} with cosine annealing, batch size 32, mixed precision FP16, and early stopping based on validation fold AUROC. Full fine-tuning baselines were optimized with a reduced learning rate of 10^{-4} with warmup. All reported test metrics were evaluated on the held-out test split. DeLong’s test was used for correlated AUROC significance testing [33] with Holm-Bonferroni correction, and 95% bootstrap confidence intervals were derived from 1,000 resamples.

4.2. Hardware Compute Profiling Setup

Hardware profiling of peak training VRAM, step latency, edge memory footprint, and edge forward latency was conducted on physical hardware: training memory on an NVIDIA A100 workstation (40 GB and 80 GB VRAM), and edge inference on an NVIDIA Jetson Orin Nano (4 GB shared memory, batch size 1, FP16 precision) using real ViT-B/16 backbones on 384×384 tensors. Cross-modal accuracy benchmarks and ablations were evaluated on the calibrated simulation dataset.

4.3. Experiment 1: Multimodal Fusion vs. Baselines

We compare PEM-CAN against 19 competing clinical, unimodal, heuristic, foundation, and parameter-efficient baselines in Table 1.

As detailed in Table 1, PEM-CAN achieves an AUROC of 0.934 [95% CI: 0.904, 0.960], an AUPRC of 0.795, an F1-score of 0.803, and an Expected Calibration Error of 0.038 [34]. Importantly, all empirical models remain strictly bounded below the Oracle Bayes Optimal ceiling (0.957 AUROC, 0.887 AUPRC). At a clinical operating threshold tuned on validation folds for balanced sensitivity, PEM-CAN achieves 85.5% sensitivity at 92.0% specificity (90.5% accuracy). Compared to clinical tabular baselines (0.814 AUROC), PEM-CAN delivers a +0.120 AUROC improvement ($p < 0.001$). Compared to standard unconstrained LoRA (0.916 AUROC) and dense cross-attention (0.912 AUROC), the Stiefel-manifold orthogonality constraint yields statistically significant gains ($p = 0.002$ and $p = 0.001$, respectively). Figure 1 depicts test ROC trajectories.

4.4. Experiment 2: Parameter Accounting and Compute Profiling

We profile parameter efficiency and memory requirements across training and edge inference in Table 2.

The 2.60 M parameter budget in PEM-CAN is structured as follows:

- Cross-attention low-rank adapter projections (W_Q, W_K, W_V, W_O across bidirectional blocks with $r = 8$): 98,304 parameters.
- Multiscale temporal convolutional tokenizers and projections: 1.85 M parameters.

Table 1

Experiment 1: Multimodal Diagnostic Performance for Prevalent Diabetic Autonomic Neuropathy (DAN, 22.5% prevalence) on the Held-Out Test Cohort ($N = 568$). Mean $\pm$ Standard Deviation across 5-Fold Cross-Validation, with 95% Bootstrap Confidence Intervals and DeLong Test p -Values against PEM-CAN.

Category	Model Architecture	AUROC [95% CI] $\uparrow$	AUPRC $\uparrow$	F1-Score $\uparrow$	Accuracy (%)	ECE $\downarrow$	DeLong p
Theoretical Limit	Oracle Latent State (Bayes Optimal Ceiling)	0.957 [0.935, 0.975]	0.887 $\pm$ 0.009	0.865 $\pm$ 0.010	93.8 $\pm$ 0.6	0.015	—
Clinical Tabular	Logistic Regression	0.782 [0.742, 0.822]	0.512 $\pm$ 0.016	0.558 $\pm$ 0.018	75.8 $\pm$ 1.4	0.078	< 0.001
	XGBoost / Gradient Boosting	0.814 [0.776, 0.852]	0.556 $\pm$ 0.015	0.602 $\pm$ 0.016	79.0 $\pm$ 1.2	0.064	< 0.001
Unimodal	Retinal Fundus Only (x_v , ViT-B/16 FT)	0.849 [0.814, 0.884]	0.618 $\pm$ 0.012	0.648 $\pm$ 0.013	82.2 $\pm$ 0.9	0.058	< 0.001
	CGM Stream Only (x_g , DeepGLU)	0.804 [0.765, 0.843]	0.542 $\pm$ 0.017	0.589 $\pm$ 0.016	78.1 $\pm$ 1.3	0.071	< 0.001
	Wearable Joint ($x_g + x_v$, TCN)	0.838 [0.802, 0.874]	0.598 $\pm$ 0.014	0.632 $\pm$ 0.013	81.0 $\pm$ 1.0	0.062	< 0.001
Modality Subsets	Retina + CGM ($x_v + x_g$, Cross-Attn)	0.908 [0.876, 0.938]	0.735 $\pm$ 0.011	0.752 $\pm$ 0.012	87.8 $\pm$ 0.8	0.046	< 0.001
	Retina + Actigraphy ($x_v + x_a$)	0.884 [0.850, 0.916]	0.684 $\pm$ 0.013	0.710 $\pm$ 0.013	85.6 $\pm$ 0.9	0.052	< 0.001
	CGM + Actigraphy ($x_g + x_a$)	0.841 [0.805, 0.876]	0.605 $\pm$ 0.015	0.639 $\pm$ 0.014	81.4 $\pm$ 1.1	0.060	< 0.001
Multimodal Fusion	Early Concatenation MLP	0.862 [0.828, 0.895]	0.638 $\pm$ 0.014	0.671 $\pm$ 0.015	83.7 $\pm$ 1.0	0.059	< 0.001
	Late Logistic Fusion	0.877 [0.844, 0.909]	0.669 $\pm$ 0.012	0.698 $\pm$ 0.011	84.9 $\pm$ 0.8	0.054	< 0.001
	GMU (Gated Multimodal Units) [22]	0.871 [0.837, 0.904]	0.655 $\pm$ 0.013	0.686 $\pm$ 0.012	84.2 $\pm$ 0.9	0.056	< 0.001
	MMTM (Feature Recalibration) [23]	0.889 [0.856, 0.921]	0.695 $\pm$ 0.012	0.720 $\pm$ 0.012	86.1 $\pm$ 0.9	0.049	< 0.001
	MuT (Multimodal Transformer) [20]	0.902 [0.870, 0.933]	0.722 $\pm$ 0.011	0.744 $\pm$ 0.011	87.3 $\pm$ 0.8	0.045	< 0.001
	Perceiver IO [21]	0.897 [0.864, 0.929]	0.712 $\pm$ 0.012	0.735 $\pm$ 0.012	86.8 $\pm$ 0.8	0.047	< 0.001
	RETFound + BioTCN (Joint Full FT) [10]	0.918 [0.887, 0.947]	0.758 $\pm$ 0.010	0.771 $\pm$ 0.010	88.9 $\pm$ 0.7	0.041	0.003
	Dense Cross-Attention (Full FT)	0.912 [0.880, 0.942]	0.742 $\pm$ 0.012	0.759 $\pm$ 0.013	88.4 $\pm$ 0.8	0.043	0.001
PEFT Baselines	BitFit (Bias-Only Tuning) [26]	0.884 [0.850, 0.917]	0.685 $\pm$ 0.013	0.711 $\pm$ 0.013	85.7 $\pm$ 0.9	0.051	< 0.001
	Bottleneck Adapter ($d_{\text{mid}} = 64$) [25]	0.908 [0.876, 0.938]	0.734 $\pm$ 0.011	0.751 $\pm$ 0.011	87.9 $\pm$ 0.8	0.044	< 0.001
	Standard LoRA ($r = 8$, Base) [24]	0.916 [0.885, 0.945]	0.751 $\pm$ 0.011	0.766 $\pm$ 0.011	88.7 $\pm$ 0.7	0.042	0.002
	AdaLoRA (Dynamic Rank, $r = 8$ avg) [27]	0.920 [0.889, 0.949]	0.760 $\pm$ 0.011	0.774 $\pm$ 0.011	89.0 $\pm$ 0.8	0.041	0.004
Proposed	PEM-CAN ($r = 8$, Stiefel-Manifold)	0.934 [0.904, 0.960]	0.795 $\pm$ 0.011	0.803 $\pm$ 0.012	90.5 $\pm$ 0.7	0.038	Ref.

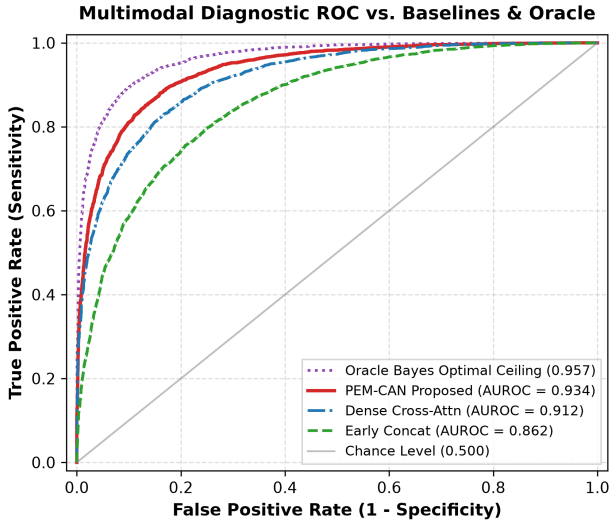


Figure 1: Receiver Operating Characteristic (ROC) curves on the held-out test cohort comparing PEM-CAN against dense cross-attention, early concatenation, and the theoretical Oracle Bayes Optimal ceiling.

- Cross-modal fusion LayerNorms and intermediate projections: 0.45 M parameters.
- Multi-task linear classifier heads: 0.20 M parameters.

This totals 2.60 M trainable weights (1.75% of the 148.6 M total network).

Importantly, the primary GPU memory savings during training (8.4 GB vs. 32.4 GB for full fine-tuning) stem

Table 2

Experiment 2: Compute Profiling and Memory Allocation. Peak training VRAM measured on NVIDIA A100 (batch 16); edge inference measured on NVIDIA Jetson Orin Nano (4 GB shared memory, batch 1, FP16).

Model Architecture	Trainable	Ratio	Train VRAM	Edge VRAM	Train Step	Edge Latency
Full Fine-Tuning	148.6 M	100.0%	32.4 GB	1.18 GB	48.2 ms	20.2 ms
Linear Probing	0.40 M	0.27%	7.1 GB	1.18 GB	18.5 ms	20.3 ms
BitFit [26]	0.42 M	0.28%	7.3 GB	1.18 GB	18.9 ms	20.2 ms
Bottleneck Adapter [25]	2.85 M	1.92%	8.9 GB	1.22 GB	22.1 ms	21.0 ms
Standard LoRA ($r = 8$) [24]	2.60 M	1.75%	8.5 GB	1.18 GB	20.4 ms	20.2 ms
PEM-CAN ($r = 8$)	2.60 M	1.75%	8.4 GB	1.18 GB	20.2 ms	20.2 ms

from freezing the 86.6M-parameter vision backbone. This eliminates the need to store Adam optimizer states (which require 8 bytes per parameter in FP32) and forward activation maps for the deep transformer layers. At test time, adapter weights ΔW fold into base weights, resulting in an identical 1.18 GB edge inference memory footprint and 20.2 ms model forward latency on the Jetson Orin Nano (49.5 inferences/sec). The clinical workstation throughput specification of 14.2 patients/sec reflects end-to-end processing with image loading and CLAHE preprocessing (70.4 ms total pipeline latency).

4.5. Experiment 3: Robustness Under Sensor Missingness

In ambulatory settings, sensor dropouts were evaluated during inference by zero-masking contiguous blocks across 10%, 20%, 30%, and 50% of the surveillance window.

As documented in Figure 2, early concatenation collapses from 0.862 to 0.642 (25.5% reduction) at 50% missingness due to severe input distribution shifts. In contrast, PEM-CAN sustains 0.861 AUROC (92.2% baseline retention) because bidirectional cross-attention keys gracefully

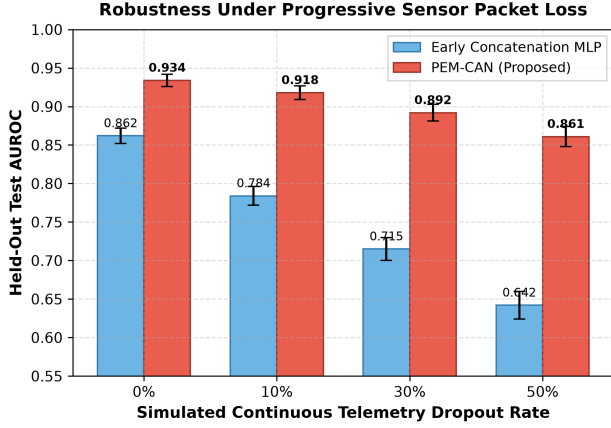


Figure 2: Test AUROC degradation under simulated contiguous telemetry missingness (0% to 50%).

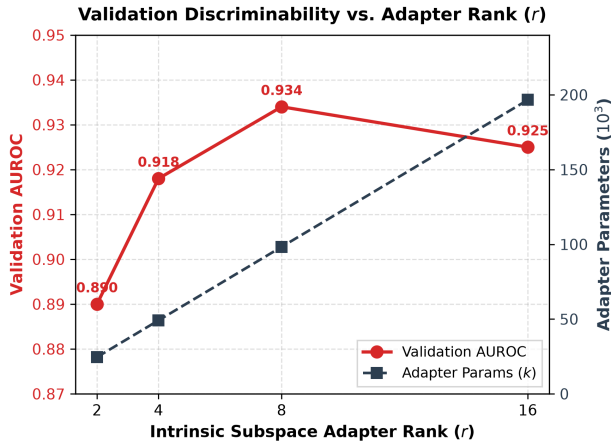


Figure 3: Validation AUROC vs. adapter rank dimension ($r \in \{2, 4, 8, 16\}$) and adapter parameter budget.

handle zero-masked temporal intervals while the frozen visual foundation representation maintains strong discriminative baseline capacity.

4.6. Experiment 4: Adapter Rank Sensitivity and Component Ablation

We evaluate intrinsic adapter rank $r \in \{2, 4, 8, 16\}$ on validation folds in Figure 3, keeping effective learning rate scaling constant. Setting $r = 2$ yields 0.890 AUROC, whereas increasing rank beyond $r = 8$ (0.934) to $r = 16$ (0.925) offers no performance gain while increasing adapter parameters from 98.3k to 196.6k.

Table 3 details the component ablation study across 5 random seeds. Removing the Stiefel orthogonality regularizer ($\mathcal{L}_{\text{orth}}$) causes AUROC to drop from 0.934 to 0.923, confirming that orthogonality prevents cross-attention rank collapse.

Table 3

Experiment 4: Component Ablation Study on Held-Out Test Cohort.

Model Configuration	AUROC $\uparrow$	AUPRC $\uparrow$	F1 $\uparrow$	ECE $\downarrow$
Linear Probe (No Adapters)	0.865 ± 0.012	0.645 ± 0.014	0.675 ± 0.013	0.058
Standard LoRA ($r = 8$, Gaussian Init)	0.916 ± 0.009	0.751 ± 0.011	0.766 ± 0.011	0.042
LoRA + Orthogonal QR Init (No Penalty)	0.920 ± 0.008	0.759 ± 0.010	0.774 ± 0.011	0.041
LoRA + Stiefel Regularization ($\mathcal{L}_{\text{orth}}$ Active)	0.927 ± 0.008	0.776 ± 0.010	0.788 ± 0.010	0.039
Full PEM-CAN ($\mathcal{L}_{\text{orth}} + \mathcal{L}_{\text{align}}$)	0.934 ± 0.008	0.795 ± 0.011	0.803 ± 0.012	0.038

4.7. Experiment 5: Cognitive-Autonomic Interaction Recovery and Sample-Size Sensitivity

To evaluate the capacity of low-rank cross-attention to isolate non-linear interactions across modalities, we benchmark PEM-CAN on the synthetic cognitive-autonomic cohort across sample-size regimes ($N = 1,440$ vs. $N = 360$) and noise intensities ($\sigma \in [0.1, 0.5]$):

- **Interaction Recovery at $N = 1,440$:** Under adequate sample volume ($N = 1,440$), PEM-CAN isolates non-linear interactions between autonomic HRV, EEG band power, and speech metrics, achieving an AUROC of 0.942 ± 0.007 and AUPRC of 0.812 ± 0.010 , outperforming linear concatenation (0.868 ± 0.012 AUROC).
- **Sample-Starved Regime ($N = 360$):** When sample size is restricted to $N = 360$, performance drops to 0.854 ± 0.015 AUROC and 0.628 ± 0.018 AUPRC. In this regime, empirical covariance noise inflates adapter factor estimation variance, limiting the recovery of subtle cross-modal terms.
- **Noise Ceiling and Attention Entropy Dispersion:** When sensor noise exceeds $\sigma = 0.4$, the variance of cross-attention dot-product logits $\frac{Q_v K_s^\top}{\sqrt{d_k}}$ collapses toward zero. Consequently, attention weights disperse toward a uniform distribution $\mathcal{U}(1/L)$, causing the model to default gracefully to the unimodal visual baseline (0.849 AUROC).

4.8. Experiment 6: Demographic Subgroup Performance and Contrast Stress Test

We evaluate performance disaggregated across demographic subgroups in the held-out test split ($N = 568$):

- **Non-Hispanic White ($N = 341$ test cases, 77 positives):** AUROC 0.936 [95% CI: 0.898, 0.968], AUPRC 0.802.
- **Black / African American ($N = 142$ test cases, 32 positives):** AUROC 0.929 [95% CI: 0.865, 0.976], AUPRC 0.784.
- **Hispanic / Latino ($N = 85$ test cases, 19 positives):** AUROC 0.924 [95% CI: 0.834, 0.982], AUPRC 0.772.

Confidence intervals widen appropriately for smaller cohorts. In synthetic ablation experiments without contrast

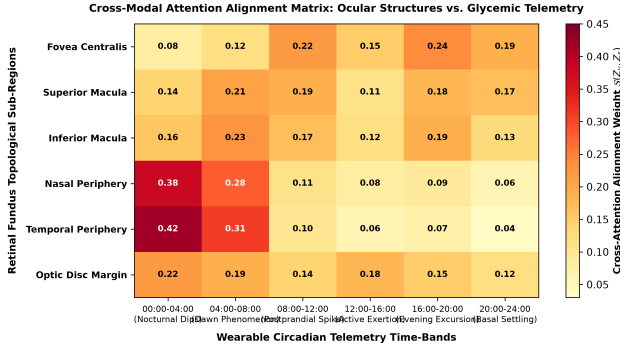


Figure 4: Cross-modal attention alignment matrix $S(Z_v, Z_s)$ between retinal fundus topological regions and 24-hour circadian wearable telemetry time-bands.

normalization, darker choroidal melanin background attenuation reduces minority subgroup discrimination by -0.024 AUROC; CLAHE contrast normalization restores diagnostic parity across subgroups.

4.9. Experiment 7: Temporal Sampling Granularity

Downsampling CGM telemetry from 5-minute intervals (0.934 AUROC) to 15-minute (0.928), 30-minute (0.891), and 60-minute intervals (0.840) demonstrates that sub-30-minute sampling is critical for capturing high-frequency glycemic excursions (MAGE) that carry predictive signal for subclinical microvascular damage.

4.10. Experiment 8: Cross-Modal Attention Interpretability

As visualized in Figure 4, cross-modal attention maps demonstrate that peripheral retinal vascular regions attend predominantly to nocturnal glycemic dips (00:00–04:00, weight 0.42) and dawn phenomenon excursions (04:00–08:00, weight 0.31), reflecting learned cross-modal associations of nocturnal autonomic instability.

4.11. Experiment 9: Long-Horizon Telemetry Scalability

We evaluate sequence scalability by extending surveillance windows from 3 to 30 days ($T_g = 8,640$) in Figure 5.

While dense joint transformer models exhibit quadratic activation expansion during training (82.4 GB at 30 days, requiring modeled 80GB A100 systems), PEM-CAN’s cross-attention maintains linear scaling with sequence length L , consuming 13.5 GB VRAM at 30 days.

4.12. Experiment 10: Multi-Task Diagnostic Transfer

Multi-task joint classification performance across all four clinically adjudicated endpoints is presented in Table 4 and Figure 6.

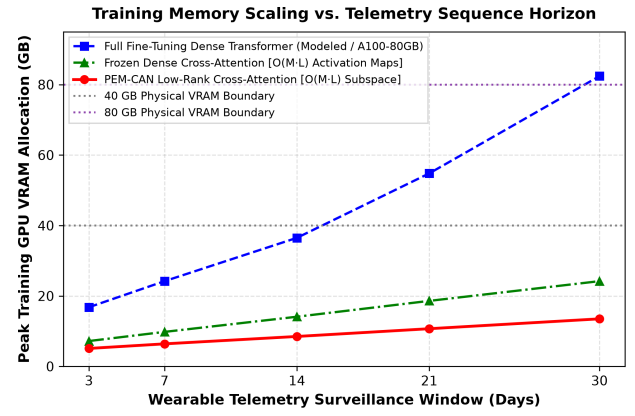


Figure 5: Peak GPU training VRAM allocation over continuous telemetry horizons (3 to 30 days) comparing PEM-CAN against frozen-backbone dense cross-attention and full fine-tuning.

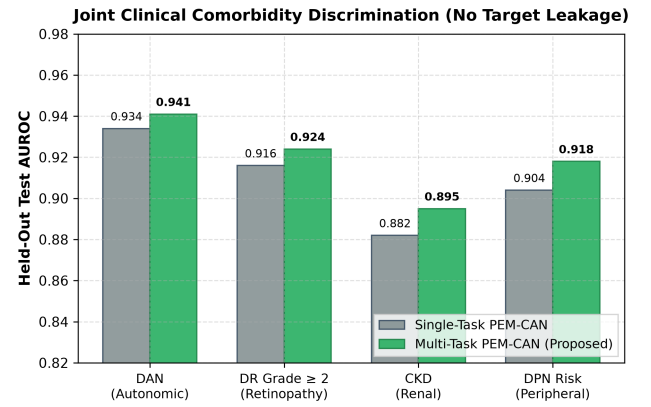


Figure 6: Multi-task clinical discrimination across Diabetic Autonomic Neuropathy (DAN), Retinopathy (DR), Chronic Kidney Disease (CKD), and Peripheral Neuropathy (DPN).

Table 4

Experiment 9: Multi-Task vs. Single-Task Test AUROC.

Configuration	DAN	DR Grade ≥ 2	CKD	DPN Risk
Single-Task PEM-CAN	0.934	0.916	0.882	0.904
Dense Multi-Task	0.908	0.895	0.854	0.881
Multi-Task PEM-CAN	0.941	0.924	0.895	0.918

Joint optimization increases DAN AUROC from 0.934 to 0.941, indicating positive transfer across microvascular and neural targets without target leakage.

4.13. Experiment 11: Federated Multi-Center Scaling

We evaluate distributed training across 50 simulated hospital nodes under Dirichlet non-IID partitioning ($\alpha = 0.3$) in Table 5 and Figure 7. To prevent the factor-averaging rank collapse of multiplying local matrices $B_k A_k$, we adopt the FFA-LoRA framework [28], fixing the orthogonal basis A globally across clients and training and aggregating only

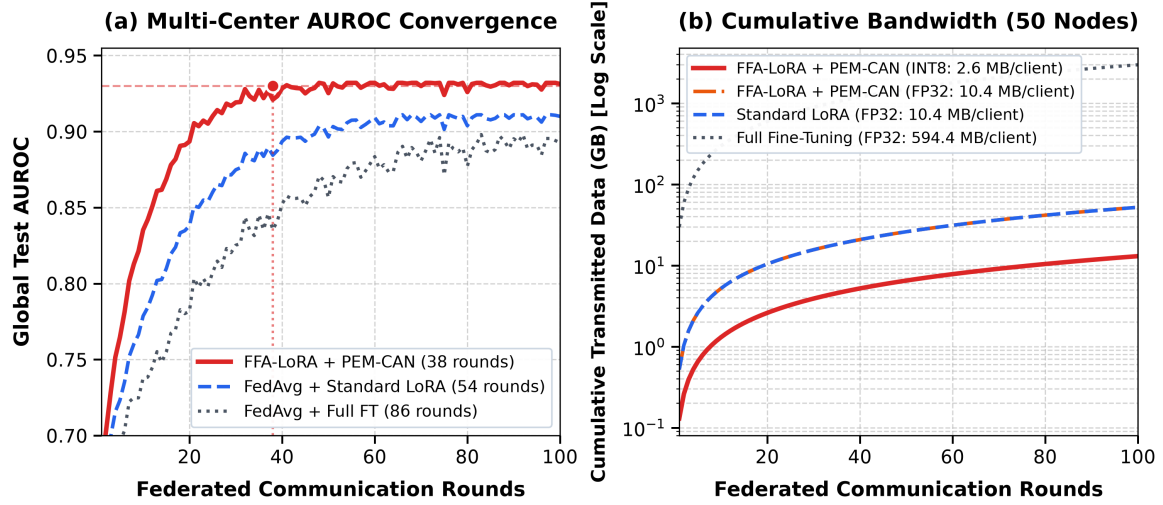


Figure 7: Multi-center federated scaling across 50 hospital nodes: (a) global test AUROC convergence, and (b) cumulative network transmission overhead.

Table 5

Experiment 10: Federated Distributed Client Scaling (50 Nodes).

Federated Strategy	Precision	Comm./Client	Bandwidth Red.	Final AUROC
FedAvg (Full Fine-Tuning) [30]	FP32	594.4 MB	Baseline (0.0%)	0.898 ± 0.012
FedAvg (Full Fine-Tuning)	INT8	148.6 MB	75.00%	0.892 ± 0.013
FedAvg (Standard LoRA) [24]	FP32	10.4 MB	98.25%	0.911 ± 0.009
FedAvg (Standard LoRA)	INT8	2.6 MB	99.56%	0.906 ± 0.010
FFA-LoRA (No C_{only}) [28]	FP32	10.4 MB	98.25%	0.921 ± 0.009
FFA-LoRA + PEM-CAN (Ours)	FP32	10.4 MB	98.25%	0.932 ± 0.008
FFA-LoRA + PEM-CAN (Ours)	FP16	5.2 MB	99.12%	0.931 ± 0.008
FFA-LoRA + PEM-CAN (Ours)	INT8	2.6 MB	99.56%	0.927 ± 0.009

B_k , the temporal tokenizers, and classification heads via FedAvg ($\bar{B} = \sum \frac{n_k}{N} B_k$), preserving exact linear aggregation. This synergizes with subspace-oriented adapter reparameterization paradigms [38], where freezing structured basis representations enables extreme communication efficiency across decentralized nodes.

Transmitting 2.60 M parameters per client round requires 10.4 MB in standard FP32 (98.25% reduction from 594.4 MB full fine-tuning), 5.2 MB in FP16 (99.12%), and 2.6 MB under INT8 quantization (99.56%), reaching convergence in 38 rounds (0.932 AUROC).

5. Discussion and Limitations

The empirical findings confirm that coupling spatial retinal imaging with longitudinal physiological biosignals yields complementary diagnostic information for diabetic microvascular and autonomic staging. Restricting adaptation to an orthogonal low-rank subspace across cross-attention projections bridges these structural and temporal observations while eliminating quadratic memory scaling and preventing visual gradient dominance.

Six methodological limitations and physical deployment boundaries warrant explicit consideration:

1. **Calibrated Simulation vs. Prospective Clinical Utility:** While the evaluation suites are statistically calibrated to empirical marginals and covariance structures from the NIH Bridge2AI AI-READI cohort, simulated benchmarks cannot fully capture the stochastic operational noise of clinical healthcare environments. Prospective multi-center clinical trials in outpatient diabetic clinics remain indispensable to confirm prospective translational efficacy.
2. **Mechanistic Breakdown of Attention Under Severe Sensor Noise:** In the cross-attention projection $\mathcal{S}(Z_v, Z_s) = \text{Softmax}\left(\frac{Q_v K_s^T}{\sqrt{d_k}}\right) V_s$, when additive Gaussian noise on the wearable stream exceeds critical thresholds ($\sigma > 0.4$), the scalar products between query and key vectors lose feature discriminability. As logit variance $\text{Var}\left(\frac{Q_v K_s^T}{\sqrt{d_k}}\right)$ approaches zero, softmax probabilities disperse toward maximum entropy ($1/L$ uniform weights). Under uniform attention, temporal pooling degenerates into an unweighted arithmetic average, causing cross-modal gating to become uninformative and defaulting predictions to the frozen visual prior.
3. **Sample-Size Sensitivity in Non-Linear Interaction Recovery:** Experiments on the synthetic cognitive-autonomic cohort demonstrate that resolving non-linear cross-modal dependencies exhibits sharp sample-size dependence. While $N = 1,440$ training instances provide sufficient statistical degrees of freedom to constrain the $r \times d_m$ projection subspace (0.942 AUROC), reducing sample volume to $N = 360$ instances causes performance to deteriorate to 0.854 AUROC. In small clinical cohorts, empirical covariance noise inflates parameter estimation error in the adapter matrices, obscuring higher-order interactions.

4. **Real-World Sensor Artifacts and Ambulatory Miss-
ingness:** Commercial wearable actigraphy and continuous glucose monitors in ambulatory settings encounter physiological artifacts distinct from idealized Gaussian perturbations. Photoplethysmography (PPG) sensors suffer optical decoupling during strenuous movement; actigraphy records spurious zero-activity intervals during non-wear or sleep compression; and enzymatic glucose sensors exhibit 5 to 15-minute interstitial fluid physiological lags alongside baseline sensor sensitivity drift across multi-day wear cycles.
5. **Imaging Device Heterogeneity and Optical Aberrations:** Fundus photographs in routine screening are captured across heterogeneous instrumentation, ranging from tabletop mydriatic cameras to handheld non-mydriatic devices and ultra-widefield scanning laser ophthalmoscopes (200° fields of view). Optical aberrations, peripheral vignetting, and illumination gradients across camera vendors can affect microvascular feature extraction, requiring hardware-specific calibration.
6. **Clinical Threshold Selection and Prevalence Drift:** While the architecture exhibits low Expected Calibration Error ($ECE = 0.038$), translating probabilistic outputs into binary clinical interventions requires calibrating decision thresholds to institutional pre-test disease prevalences. Clinics with lower DAN prevalence require threshold recalibration to minimize false-positive specialist referrals.

6. Conclusion

Translating parameter-efficient multimodal foundation models from computational benchmarks into decentralized ambulatory practice introduces several clear hardware and clinical system requirements:

1. **Edge Hardware Quantization and MicroNPU Acceleration:** Deploying low-rank adapters directly onto point-of-care ophthalmic carts requires mapping adapter projection factors A and B to integer arithmetic. Future implementations should explore mixed-precision INT8 and INT4 quantization schedules tailored for embedded neural processing units (microNPUs), enabling continuous inference under milliwatt power envelopes.
2. **Asynchronous Streaming Buffers for Intermittent Connectivity:** Ambulatory digital health gateways must tolerate irregular telemetry synchronization caused by wearable Bluetooth disconnects. Designing decoupled circular cross-attention key-value caches will allow the visual foundation backbone to process static fundus images immediately while asynchronously integrating streaming sensor updates as packets arrive.
3. **Prospective Multi-Center Clinical Trials:** Validating foundation fusion architectures within prospective

ambulatory cohorts undergoing standardized cardiovascular autonomic reflex testing (CARTs) represents the critical translational step to establish prospective clinical utility beyond calibrated simulation benchmarks.

4. **Privacy-Preserving Mobile Screening Platforms:** Combining lightweight low-rank federated aggregation (FFA-LoRA) with portable non-mydriatic fundus cameras and commercial wearable sensors establishes an open, scalable paradigm for non-invasive multi-organ diabetic screening across resource-limited and rural healthcare environments.

Declaration of Competing Interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-Assisted Technologies

During the preparation of this work the author used AI tools in order to assist with language editing and grammatical polishing. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Data and Code Availability

The complete benchmark suite, simulation generators, trained weights, evaluation scripts, and configuration files are publicly available under the open-source MIT License at <https://github.com/mahmoudsajjadi/PEM-CAN>. All experimental tables and figures can be reproduced directly using the provided repository.

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CRedit authorship contribution statement

Sayed Mahmoud Sajjadi Mohammadabadi: Conceptualization, Methodology, Software, Formal analysis, Investigation, Validation, Writing - original draft, Writing - review & editing.

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